

Online Retail Sales Analysis using SQL

Analysis Results & Business Insights

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Tools Used

SQL Server • SQL • GitHub

GitHub Repository

<https://github.com/buildwithaashi/Retail-Sales-Analytics>

SQL Data Analytics Project

Q1. Total Revenue Generated



The image shows a screenshot of a software application with two tabs: 'Results' and 'Messages'. The 'Results' tab is active, displaying a table. The table has two columns. The first column contains the number '1'. The second column contains the text 'Total_Revenue' in the header row and the value '1320629.44' in the data row. The value '1320629.44' is highlighted with a dashed black border.

	Total_Revenue
1	1320629.44

Analysis Summary

Calculated the total revenue generated from all valid retail transactions after cleaning. This KPI provides an overview of the company's overall sales performance.

Q2. Revenue by Country

	country	Revenue
1	United Kingdom	1157904.11
2	EIRE	30288.4
3	Netherlands	27200.86
4	Germany	23094.4
5	France	21943.93
6	Spain	10056.52
7	Australia	9665.88
8	Japan	7705.07
9	Portugal	5811.75
10	Switzerland	4909.55
11	Sweden	4381.86
12	Norway	3787.12
13	Belgium	2686.67
14	Italy	2420.16
15	Cyprus	2138.32
16	Lithuania	1881.08

Analysis Summary

Compared revenue generated across countries. The United Kingdom contributed the highest revenue, highlighting it as the primary revenue-generating market.

Q4 & Q5. Product Performance

	Description	Revenue
1	MEDIUM CERAMIC TOP STORAGE JAR	77183.6
2	REGENCY CAKESTAND 3 TIER	36729.61
3	DOTCOM POSTAGE	34177.86
4	WHITE HANGING HEART T-LIGHT HOLDER	24558.54
5	AMAZON FEE	13541.33
6	CHILLI LIGHTS	12825.46
7	RED WOOLLY HOTTIE WHITE HEART.	9430.87
8	PAPER CHAIN KIT 50'S CHRISTMAS	9369.12
9	WHITE SKULL HOT WATER BOTTLE	8889.81
10	HEART OF WICKER LARGE	8178.24

Fig 2:-For Q4

	Description	Quantity_Sold
1	MEDIUM CERAMIC TOP STORAGE JAR	74215
2	WHITE HANGING HEART T-LIGHT HOLDER	8720
3	PACK OF 72 RETROSPOT CAKE CASES	6665
4	WORLD WAR 2 GLIDERS ASSTD DESIGNS	6109
5	HAND WARMER BABUSHKA DESIGN	4181
6	JUMBO BAG RED RETROSPOT	4118
7	HEART OF WICKER SMALL	4049
8	SMALL POPCORN HOLDER	3993
9	MINI PAINT SET VINTAGE	3615
10	ASSORTED COLOUR BIRD ORNAMENT	3374

Fig 2:-For Q5

Analysis Summary

Identified products generating the highest revenue and the highest sales volume. High-revenue products were not always the highest-selling by quantity, indicating differences between pricing and demand.

Q6. Products Appearing in the Highest Number of Invoices

	Description	Unique_Invoices
1	WHITE HANGING HEART T-LIGHT HOLDER	338
2	REGENCY CAKESTAND 3 TIER	247
3	HEART OF WICKER SMALL	217
4	HAND WARMER BABUSHKA DESIGN	205
5	SCOTTIE DOG HOT WATER BOTTLE	203
6	JAM MAKING SET PRINTED	200
7	JUMBO BAG RED RETROSPOT	198
8	CHOCOLATE HOT WATER BOTTLE	175
9	PAPER CHAIN KIT 50'S CHRISTMAS	173
10	HEART OF WICKER LARGE	169

Analysis Summary

Measured how frequently products appeared across invoices. Frequently purchased products represent consistently demanded items.

Q7, Q8 & Q9. Customer and Geographic Analysis

	Customer_id	Revenue		customer_id	Unique_Invoices
1	12346	77183.6	1	12748	39
2	18102	27834.61	2	17850	34
3	14646	27008.26	3	14911	16
4	15749	22998.4	4	15061	16
5	15061	22107.61	5	15311	16
6	17450	20649.04	6	14606	13
7	14156	17096.92	7	13089	13
8	16029	14820.15	8	17841	12
9	14911	11218.36	9	13777	11
10	17511	10573.22	10	16029	10

	Country	Total_Customers
1	United Kingdom	1029
2	France	30
3	Germany	29
4	Australia	7
5	Spain	7
6	Portugal	7
7	Belgium	5
8	Switzerland	3
9	Italy	3
10	EIRE	3

Analysis Summary

Identified the highest-spending customers, customers placing the most orders, and countries with the highest number of unique customers. These findings support customer retention and market expansion strategies.

Q10. Daily Revenue Trend

	Invoice_date	Revenue
1	2010-12-01	58960.79
2	2010-12-02	47748.38
3	2010-12-03	46943.71
4	2010-12-05	31774.95
5	2010-12-06	54830.46
6	2010-12-07	99618.2
7	2010-12-08	45389.98
8	2010-12-09	53586.18
9	2010-12-10	59182.92
10	2010-12-12	17329.07
11	2010-12-13	38006.71
12	2010-12-14	45254.73
13	2010-12-15	30444.62
14	2010-12-16	49352.9
15	2010-12-17	45418.33
16	2010-12-19	7534.91

Analysis Summary

Analyzed revenue over time. Sales remained relatively stable with normal day-to-day fluctuations and no strong long-term trend.

Q11 & Q12. Above-Average Product Analysis

	description	Average_Unit_Price
1	AMAZON FEE	13541.330078125
2	DOTCOM POSTAGE	371.498478596625
3	LOVE SEAT ANTIQUE WHITE METAL	175
4	VINTAGE RED KITCHEN CABINET	167.5
5	REGENCY MIRROR WITH SHUTTERS	165
6	RUSTIC SEVENTEEN DRAWER SIDEBORD	160
7	VINTAGE BLUE KITCHEN CABINET	125
8	CHEST NATURAL WOOD 20 DRAWERS	117.5
9	VINTAGE POST OFFICE CABINET	66.3599987030029
10	SCHOOL DESK AND CHAIR	65
11	CARRIAGE	58.33333333333333
12	Manual	54.9856830076738
13	SET/4 WHITE RETRO STORAGE CUBES	51.1518190557306
14	HALL CABINET WITH 3 DRAWERS	44.1966667175293
15	PINK PAINTED KASHMIRI CHAIR	42.9500007629395

	description	Total_Revenue
1	MEDIUM CERAMIC TOP STORAGE JAR	77183.5971689224
2	REGENCY CAKESTAND 3 TIER	36729.609629631
3	DOTCOM POSTAGE	34177.8600308895
4	WHITE HANGING HEART T-LIGHT HOLDER	24558.539781332
5	AMAZON FEE	13541.330078125
6	CHILLI LIGHTS	12825.4599742889
7	RED WOOLLY HOTTIE WHITE HEART.	9430.87022662163
8	PAPER CHAIN KIT 50'S CHRISTMAS	9369.11993455887
9	WHITE SKULL HOT WATER BOTTLE	8889.81018686295
10	HEART OF WICKER LARGE	8178.2400829792
11	HOT WATER BOTTLE TEA AND SYMPATHY	8136.17985892296
12	CHOCOLATE HOT WATER BOTTLE	8048.46995925903
13	POSTAGE	7927.43000030518
14	WOOD BLACK BOARD ANT WHITE FINISH	7807.74024009705
15	JUMBO BAG RED RETROSPOT	7612.42000675201
16	HEART OF WICKER SMALL	7509.16990363598

Analysis Summary

Identified products with above-average selling prices and products generating above-average revenue to highlight high-value products.

Business Insights & Recommendations

Key Insights

- United Kingdom generated the highest revenue.
- High-revenue products were different from the highest-selling products.
- High-spending customers differed from the customers placing the most orders.
- Daily revenue remained relatively stable across the analysis period.

Recommendations

- Focus inventory planning on consistently high-demand products.
- Review low-demand inventory for optimization
- Introduce loyalty programs for high-value customers.
- Include product cost information in future analyses to evaluate profitability.